# RAMANNA CHOWDARY UPPULURI

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## SUMMARY

Software Engineer with 2.5 years building and validating production C++ and C#/.NET software at HCL Tech; MS in Computer Science. Builds secure Java/Spring Boot REST APIs with PostgreSQL, Docker, CI/CD, and automated testing.

## EDUCATION

**Master of Science in Computer Science** | Wright State University, OH | GPA 3.40/4.00 *Sep 2024 – May 2026*
**B.E. Electronics & Communication** | Sathyabama Institute of Science & Technology | GPA 3.60/4.00 *Aug 2018 – Jun 2022*

## TECHNICAL SKILLS

**Languages:** Java, C++, C#, Python, SQL
**Frameworks & Libraries:** Spring Boot, Spring Security, Spring Data JPA, .NET, Flask, JUnit 5, Mockito, Testcontainers
**Databases:** PostgreSQL, MySQL, Microsoft SQL Server, Flyway
**Cloud & DevOps:** AWS (EC2), Docker, Docker Compose, GitHub Actions (CI/CD), Render, Git, Linux (Ubuntu)
**Tools:** IntelliJ IDEA, Visual Studio, Postman, Swagger / OpenAPI, Jira
**Concepts:** OOP, Data Structures & Algorithms, REST APIs, Multithreading, Concurrency, Unit & Integration Testing, Agile

## PROFESSIONAL EXPERIENCE

**Senior Software Engineer | HCL Tech, Andhra Pradesh, India** *Dec 2023 – Aug 2024*
*HSD64 Development Project – Client: Teradyne Inc.*
- Ported several APIs from the 32-bit HSD32 platform to HSD64, writing C++ code to support 64-bit architecture.
- Used HSD32 functionality as the reference implementation to build and verify matching 64-bit behavior.
- Resolved 20+ software/hardware integration defects via log analysis and regression testing with firmware/hardware teams.
- Built automated test suites covering 100+ functional and integration scenarios, improving test efficiency by 30%.
- Passed all changes through senior-engineer code review against HSD64 requirements before promotion.

**Software Engineer | HCL Tech, Chennai, India** *Aug 2022 – Nov 2023*
*Eagle Test Systems (ETS) Sustenance Project – Client: Teradyne Inc.*
- Diagnosed and fixed 25+ internal and customer-reported issues in production C++ and C#/.NET software.
- Validated fixes against real-time data on physical testers at US client sites (Buffalo, NY; North Reading, MA).
- Logged new issues found during validation; promoted reviewed fixes and closed tickets within sprints.

**Graduate Software Engineer Trainee | HCL Tech, Chennai, India** *Jan 2022 – Jul 2022*
- Built object-oriented C++ and C#/.NET development skills on hardware-integrated test systems.
- Supported senior engineers in debugging, system bring-up, and equipment testing; tracked tasks in Jira.

## PROJECTS

**OrderHub: Order Management REST API** | Java, Spring Boot, Spring Security, PostgreSQL, Docker | GitHub | Live Demo
- Built a layered Spring Boot REST API for products, customers, and orders with JWT authentication and role-based access.
- Implemented transactional ordering with pessimistic row locks to prevent overselling and an enforced status lifecycle.
- Modeled PostgreSQL data with Spring Data JPA and Flyway, adding pagination, validation, and global error handling.
- Wrote JUnit 5/Mockito unit tests and Testcontainers integration tests, run automatically in a GitHub Actions CI pipeline.
- Containerized with Docker, deployed on Render with Neon PostgreSQL, and documented the API with Swagger.

**Automated Test Framework for Simulated Semiconductor Equipment** | C#/.NET, C++, GitHub Actions | GitHub
- Designed a C#/.NET test automation framework simulating 5 hardware domains with 19 automated test cases.
- Implemented multithreaded parallel test execution, cutting total runtime by nearly a third.
- Built an advanced logging system, JSON/CSV result export, and fault-condition simulation to speed up defect analysis.

**Code Pathway: AI Coding Learning Platform** | Team project | Python, Flask, MySQL, AWS EC2 | GitHub
- Co-developed a web platform giving students real-time AI feedback on code (Distributed Computing course).
- Designed the MySQL database structure for users, submissions, and feedback history.
- Deployed the application on AWS EC2 and integrated the front end with the Flask back end.

## CERTIFICATIONS

**AWS Certified Cloud Practitioner** | Amazon Web Services | Expected Dec 2026